

1(a)(i)	<i>independent</i> light intensity ; <i>dependent</i> distance moved by bubble (in a set time) or time taken for the bubble to move (a set distance) ;	2
1(a)(ii)	<i>any three variables for one mark</i> <i>two correct methods for one mark each</i> 1 temperature – temperature-controlled room / heat shield / environmental chamber / incubator / cold light source / LED light ; 2 background light – dark room (with fixed light) / closed blinds / AW ; 3 same wavelength / type / colour of light – use the same bulb / lamp / AW ; 4 <i>idea of</i> air flow / wind – e.g. windows closed / doors closed / fans off or on throughout / limit movement / AW ; 5 plant / (leafy) shoot – use same, type / species / plant / shoot / age / number of leaves / area of leaves ; 6 humidity – any valid way it might be controlled ; 7 CO ₂ levels – any valid way it might be controlled ; 8 time – measure for same time / time over same distance ;	3

Question	Answer	Marks
1(b)	<i>any six from</i> 1 <i>ref. to</i> cutting / inserting, stem under water ; 2 use, petroleum jelly / silicone gel / silicone tape / AW (to make joints air tight) ; 3 <i>idea of</i> removing tube from water to introduce an air bubble ; 4 <i>ref. to</i> method of obtaining a minimum of 5 different light intensities ; 5 <i>ref. to</i> method of controlling one variable ; 6 allow (apparatus / plant) to, equilibrate / AW, before starting measurements ; 7 <i>idea of</i> setting / resetting / returning, air bubble (to start position / between measurements) ; 8 <i>ref. to</i> measuring distance moved by bubble over a set time or <i>ref. to</i> measuring time for bubble to move a set distance ; 9 take a minimum of 2 repeats at each light intensity + take a mean / identify anomalies ; 10 low / medium, risk investigation or cutting stem + cut away from your hand (with scalpel) / cut with secateurs or scissors ;	6
1(c)(i)	30.0 (%) ; $(3.03 - 2.12) / 3.03$ or $(1 - 2.12 / 3.03) \times 100$	2

Question	Answer	Marks
1(c)(ii)	<i>any two from</i> 1 <i>idea of</i> only two, concentrations of CO₂/ light intensities, measured ; 2 <i>idea of</i> only 1, type / species, of plant tested ; 3 not replicated / repeated ; 4 <i>idea that</i> laboratory conditions may not be replicated in the field ; 5 no statistical analysis ; 6 <i>idea of</i> other (stated) variable(s), not controlled / can affect transpiration ;	2

Question	Answer	Marks
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